EMPLOYEE ATTRITION ANALYTICS

A Data-Driven Investigation of Voluntary Turnover
at a Multi-Department Organization

A Data Analytics, Data Science and Business Intelligence Project

Dataset: IBM HR Analytics Employee Attrition & Performance (Kaggle)

Final Project Report

1. Executive Summary

This report presents an end-to-end analytics investigation of employee attrition using the IBM HR Analytics dataset (1,470 employees, 35 variables). The workforce's overall attrition rate stands at 16.1%, concentrated heavily among employees who work overtime (30.5% vs 10.4%), employees in their first two years of tenure (29.8%), and specific roles such as Sales Representative (39.8%).

A supervised machine-learning pipeline comparing Logistic Regression, a Decision Tree, and a Random Forest classifier found Logistic Regression to be the best-balanced model (Precision 73.7%, Recall 29.8%, F1 0.424, ROC-AUC 0.777), consistently ranking OverTime as the single strongest predictor of attrition across all three algorithms.

Eleven business insights, four KPIs, and a three-page Power BI dashboard blueprint translate these findings into a concrete, monitorable retention programme. The core recommendation is that overtime and workload management — not blanket compensation increases — represent the highest-return, lowest-cost lever available to reduce attrition.

2. Dataset Description

The dataset used is IBM's HR Analytics Employee Attrition & Performance dataset, sourced from Kaggle (kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset). It contains 1,470 employee records across 35 variables spanning demographic (Age, Gender, MaritalStatus), compensation (MonthlyIncome, DailyRate, PercentSalaryHike, StockOptionLevel), tenure (YearsAtCompany, TotalWorkingYears, YearsSinceLastPromotion), satisfaction (JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance, RelationshipSatisfaction), and organizational (Department, JobRole, JobLevel, BusinessTravel) categories, with Attrition (Yes/No) as the target variable. Full column-level definitions and the working dataset are provided in the Dataset_Overview and Data_Cleaned tabs of the accompanying workbook.

3. Dataset Justification

This dataset was selected because it satisfies every technical and analytical requirement of the brief: over 500 records (1,470), more than 8 columns (35), a genuine mix of numerical and categorical variables, embedded business metrics (MonthlyIncome, PerformanceRating, JobSatisfaction), and a binary target variable (Attrition) that is directly actionable by HR decision-makers. Full justification — business context, business problem, analytical objectives, and expected business value — is documented in the Dataset_Overview tab.

Because the native Kaggle file is unusually clean, realistic data-quality problems (missing values, duplicates, inconsistent categorical labels, incorrect data types, and outliers) were deliberately injected into a working copy and fully documented, satisfying the assignment's data-cleaning requirements without misrepresenting the original source (see Data_RawIssues and Cleaning_Log).

4. Data Cleaning and Preparation

Six categories of data-quality issues were identified and corrected in a documented Python (pandas/NumPy) pipeline, fully reproduced in the Cleaning_Log tab:

1. 40 missing values across MonthlyIncome, EnvironmentSatisfaction, Department, TotalWorkingYears, and BusinessTravel, imputed using group-wise medians (numeric) or mode (categorical).
2. 15 exact duplicate employee records, removed via de-duplication on EmployeeNumber.
3. 21 inconsistent categorical labels (e.g. “sales” / “SALES” / “Sales ”) standardized via a lookup-mapping function.
4. 10 numeric fields incorrectly typed as text (e.g. “35 years”, “$5,993”), parsed back to numeric.
5. 8 statistical outliers (e.g. Age = 202, MonthlyIncome = $999,999) identified via business-rule bounds and corrected via imputation or sign-correction.
6. 2 additional scale-corruption errors discovered during verification (percentage values auto-converted to decimal fractions), corrected via a ×100 rescaling rule.

The cleaned, feature-engineered dataset (1,470 unique employees, 47 columns including 10 engineered features such as SatisfactionIndex, IncomeBand, and TenureRatio) is provided in Data_Cleaned. AI-assisted recommendations for preventing these issues at the source system are documented at the foot of the Cleaning_Log tab.

5. Exploratory Data Analysis

Full exploratory analysis — summary statistics, missing-value verification, distribution histograms, a box-and-whisker income comparison, a correlation heatmap, category comparisons, tenure-trend analysis, grouped aggregations, and a target-variable breakdown — is provided in the EDA_Summary tab with seven supporting charts.

Headline findings: the workforce skews toward the 25–44 age band; MonthlyIncome is right-skewed with a small number of high-earner outliers visible in the box plot; and OverTime shows the strongest correlation with Attrition (r = 0.245) of any variable measured, ahead of MonthlyIncome (r = -0.158), Age (r = -0.162), and TotalWorkingYears (r = -0.167).

Segment	Attrition Rate
Overtime workers vs. non-overtime workers	30.5% vs 10.4%
Employees with 0–2 years tenure	29.8%
Sales Representative role	39.8%
Single employees vs. married employees	25.6% vs 12.4%

Employees under 30 vs. 40–49 age band	27.8% vs 9.8%

Nine explicit business questions (e.g. “Does overtime drive attrition?”, “Is attrition concentrated in early tenure?”) are answered directly with supporting evidence at the foot of the EDA_Summary tab.

6. Machine Learning Model

Three supervised classification models were trained on a stratified 80/20 train-test split (1,176 training, 294 test employees, seeded for reproducibility) using 33 engineered features: Logistic Regression (L2-regularized), a Decision Tree (CART, max depth 5), and a Random Forest (15-tree bootstrap ensemble).

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	87.1%	73.7%	29.8%	0.424	0.777
Decision Tree (max depth 5)	83.7%	—	—	0.351	0.694
Random Forest (15 trees)	85.4%	—	12.8%	—	0.792

Logistic Regression was selected as the recommended production model because it offers the best Precision/Recall balance (highest F1) while remaining fully interpretable to HR stakeholders. All three models independently rank OverTime as a top-two predictor, corroborating the EDA finding. Full methodology, the model-comparison chart, per-model feature importance, and equivalent scikit-learn code are provided in the ML_Model tab.

7. Business Insights

Eleven business insights were derived directly from the EDA and modelling outputs, each backed by a specific supporting figure and paired with a concrete recommendation (full table in Business_Insights). Highlights include:

- Overtime employees leave at nearly three times the rate of non-overtime employees.
- Attrition risk falls steadily from 29.8% in the first two years of tenure to 8.1% after eleven years.
- Sales Representatives attrit at 39.8%, far above any other role.
- Employees who leave are paid less than those who stay in every department, with the largest pay gap in Human Resources ($3,461/month).

8. KPI Analysis

Four KPIs were defined to make the analysis operationally monitorable (full detail, formulas, and targets in the KPIs tab):

KPI	Actual	Target	Status
Overall Attrition Rate	16.1%	12%	Below Target
OverTime Attrition Rate	30.5%	20%	Below Target

Early-Tenure Attrition Rate	29.8%	20%	Below Target
Retention-Model Precision	73.7%	70%	On / Above Target

Three of four KPIs are currently below target, confirming that a structured retention intervention is warranted rather than optional, while the fourth confirms the predictive model is already precise enough to operationalize.

9. Dashboard Design

A three-page Power BI dashboard blueprint (Executive Summary, Operational Analysis, Predictive Insights) is specified in full in the PowerBI_Dashboard_Spec tab, including every visual's purpose, data fields, and the specific business insight it produces, plus four global slicers (Category/JobRole, Department-as-Region, TenureBucket-as-Date, and Attrition/IncomeBand segments). Design follows an inverted-pyramid principle (headline KPIs first, detail on demand), consistent colour-encoding for Attrition status across all pages, and cross-filtering / drill-through between pages.

10. AI-Generated Insights

Pattern-scanning of the EDA and model outputs (documented in full, including derivation method and data validation, in AI_Insights) produced five further observations:

- Attrition risk decays monotonically with tenure, resembling a retention “survival curve.”
- OverTime acts as a compounding risk factor, validated by its top-ranking in two independent model families.
- Role-specific attrition (Sales Representative, Laboratory Technician) is masked by department-level aggregation.
- Overtime / workload management likely offers a lower-cost intervention than blanket compensation increases.
- Early-career, single, lower-income employees form one overlapping flight-risk cluster rather than three independent ones.

Each insight is explicitly labelled with its support level (strongly supported / moderately supported) based on whether the underlying data confirms or merely suggests the pattern.

11. Recommendations

1. Audit and rebalance workload in teams with chronic overtime — the single highest-leverage, lowest-cost intervention identified.
2. Strengthen onboarding and structured check-ins through the first two years, the highest-risk tenure window.
3. Conduct a targeted compensation review for Sales Representatives and the bottom income quartile company-wide.
4. Operationalize the Logistic Regression model as a monthly HR watch-list, prioritizing high-probability, high-value employees.
5. Implement the source-system data-quality controls proposed in Cleaning_Log (validation dropdowns, uniqueness constraints, automated range checks) to keep future analysis reliable.

6 Track the four KPIs defined in this project monthly and re-run this analysis quarterly to measure whether interventions are closing the gap to target.

12. Conclusion

This project demonstrates a complete analytics lifecycle — from data cleaning and exploratory analysis through predictive modelling, KPI definition, and dashboard design — applied to a genuine business problem with material financial stakes. The evidence consistently points to overtime and early-tenure disengagement as the two most actionable drivers of attrition, ahead of compensation and demographic factors.

With three of four KPIs currently below target, the organisation has clear, quantified room for improvement, and the Logistic Regression model provides a ready-to-use tool for prioritizing where to intervene first. Subsequent work should focus on piloting the overtime-reduction recommendation, tracking its effect on the KPI dashboard, and progressively enriching the dataset (e.g. adding hire dates and location) to support true time-series and geographic analysis in future iterations.